COMPUTER NETWORKS

Practical Viva Question Bank

Detailed questions and examiner-ready answers for all 10 experiments

Course	Computer Networks Laboratory
Department	Computer Engineering
Institution	JSPM’s Rajarshi Shahu College of Engineering, Tathawade, Pune
Coverage	Experiments 1–10 from the uploaded CN Lab Manual

How to use this PDF

Read the short answer first, then explain the reason or example in your own words. For socket and protocol experiments, practise drawing the communication flow and explaining what happens at the sender, network, and receiver. Questions are grouped by the exact experiment sequence in the lab manual.

Important accuracy note

The question bank follows the 10-experiment list in the uploaded manual. Where a question asks about practical implementation details beyond the brief viva list, the answer explains standard networking behaviour. In particular, CRC detects errors but does not itself correct them; TCP is a byte stream; UDP does not guarantee delivery; and modern Facebook traffic is normally encrypted with HTTPS.

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Total questions: 150 (15 per experiment).

Experiment 1 — Setting up a Small Network (2PC/4PC) and Resource Sharing

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. What is a computer network?

Answer: A computer network is a group of interconnected devices that exchange data and share resources such as files, printers, applications, and internet access. Devices communicate through wired or wireless links using agreed protocols.

Q2. What is a LAN?

Answer: A Local Area Network connects devices within a limited geographical area such as a room, laboratory, office, or building. It is usually privately managed and offers relatively high data rates with low delay.

Q3. What is the role of a switch in this experiment?

Answer: A Layer 2 switch connects PCs in the same Ethernet LAN. It learns source MAC addresses and forwards frames only through the port associated with the destination MAC address when known; unknown unicast and broadcast frames are flooded within the VLAN.

Q4. What is the difference between a hub and a switch?

Answer: A hub is a Layer 1 repeater that sends incoming bits to every other port. A switch works mainly at Layer 2, learns MAC addresses, and forwards frames selectively. Switches reduce unnecessary traffic and give each port its own collision domain.

Q5. Why do the PCs need unique IP addresses?

Answer: Each host needs a unique IP address within the LAN so that packets can be delivered to the correct machine. Duplicate IP addresses can cause address conflicts and unreliable connectivity.

Q6. What is a subnet mask?

Answer: A subnet mask separates the network portion from the host portion of an IPv4 address. For example, 255.255.255.0 (/24) means the first 24 bits identify the network and the last 8 bits identify hosts.

Q7. Why can devices in the same subnet communicate directly?

Answer: They determine from the IP address and subnet mask that the destination is on the local network. The sender resolves the destination IPv4 address to a MAC address using ARP and sends an Ethernet frame directly through the switch.

Q8. What is the difference between static IP and DHCP?

Answer: A static IP is configured manually and remains fixed until changed. DHCP automatically leases IP configuration such as address, subnet mask, gateway, and DNS server to clients, reducing manual configuration.

Q9. What is the purpose of the default gateway?

Answer: The default gateway is the router interface used to reach destinations outside the local subnet. It is not normally needed for communication between two hosts in the same subnet.

Q10. What does the ping command test?

Answer: Ping sends ICMP Echo Request messages and waits for Echo Reply messages. Successful replies indicate IP-level reachability and provide round-trip-time and packet-loss information; ping alone does not prove that file sharing is configured correctly.

Q11. What is ARP?

Answer: Address Resolution Protocol maps an IPv4 address to a MAC address on the local network. A host broadcasts an ARP request and the device owning the IP normally responds with its MAC address.

Q12. Which protocol is commonly used by Windows file sharing?

Answer: Windows file and printer sharing commonly uses SMB (Server Message Block), typically over TCP port 445 in modern networks. Permissions and firewall rules also affect whether a share can be accessed.

Q13. What is the difference between a workgroup and a domain?

Answer: A workgroup is a peer-based arrangement where each computer manages its own local users and resources. A domain uses centralized identity and policy management, commonly through directory services.

Q14. Why might ping work while a shared folder does not open?

Answer: IP connectivity may be working while SMB is blocked by the firewall, file sharing is disabled, permissions are insufficient, the share path is wrong, or name resolution fails. Test the share by IP and check SMB/firewall settings.

Q15. What is a MAC address?

Answer: A MAC address is a link-layer identifier assigned to a network interface. Ethernet switches use source and destination MAC addresses to learn and forward frames within a LAN.

Experiment 2 — WAN with Wired and Wireless LAN using Packet Tracer

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. What is a WAN?

Answer: A Wide Area Network connects geographically separated networks, often across cities or regions. Routers forward packets between the LANs using IP addressing and routing information.

Q2. Differentiate LAN, WLAN, and WAN.

Answer: A LAN covers a limited local area. A WLAN is a LAN that uses wireless radio communication, usually Wi-Fi. A WAN connects multiple LANs over a larger geographical area using routers and WAN links.

Q3. What is Cisco Packet Tracer?

Answer: Packet Tracer is a network simulation and learning tool used to build virtual topologies, configure routers and switches, connect end devices, and test packet flow without requiring all physical equipment.

Q4. What is the role of a router in this topology?

Answer: A router connects different IP networks and forwards packets based on its routing table. In this experiment, routers connect separate LAN segments across a WAN link.

Q5. What is the difference between a switch and a router?

Answer: A switch forwards Ethernet frames within a Layer 2 network using MAC addresses. A router forwards IP packets between different networks using IP addresses and routing tables.

Q6. Why must different LANs use different network addresses?

Answer: Each routed interface must belong to an appropriate IP subnet. If two separate routed LANs use overlapping address ranges, routing becomes ambiguous and communication may fail.

Q7. What is static routing?

Answer: Static routing is a route manually configured by an administrator. It tells a router which next hop or exit interface to use to reach a remote network.

Q8. What is dynamic routing?

Answer: Dynamic routing uses routing protocols to exchange reachability information and update routing tables automatically. Examples include RIP, OSPF, and EIGRP.

Q9. What is a WAN serial link?

Answer: A serial WAN link is a point-to-point connection between routers in a simulation or legacy WAN environment. Its two ends need compatible addressing and link settings; a DCE end may need a clock rate in Packet Tracer.

Q10. What is DCE and DTE?

Answer: DCE (Data Communications Equipment) provides clocking on a synchronous serial link. DTE (Data Terminal Equipment) receives the clock. In Packet Tracer, the DCE side is commonly configured with a clock rate.

Q11. What is an access point?

Answer: A wireless access point bridges wireless stations to a wired network. It advertises a wireless network name (SSID) and coordinates Wi-Fi communication.

Q12. What is an SSID?

Answer: SSID stands for Service Set Identifier. It is the human-readable name used to identify a Wi-Fi network.

Q13. Why use WPA2 or WPA3 security?

Answer: Wi-Fi security protects wireless communication using authentication and encryption. WPA2-Personal commonly uses AES-CCMP; WPA3 provides stronger modern protections where supported. Open Wi-Fi has no equivalent access protection.

Q14. How do you verify end-to-end connectivity in Packet Tracer?

Answer: Assign correct IP addresses and gateways, configure router interfaces and routes, ensure interfaces are enabled, then ping between hosts on different LANs. Packet Tracer Simulation mode can show how packets traverse devices.

Q15. What does the command 'no shutdown' do on a Cisco router interface?

Answer: It administratively enables the interface. A correctly configured interface can still remain down if the cable, peer interface, or physical/link settings are incorrect.

Experiment 3 — Error Detection for ASCII Codes using CRC

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. What is error detection?

Answer: Error detection is the process of determining whether data changed during transmission or storage. Techniques add redundant information so the receiver can test whether the received data is likely corrupted.

Q2. What is CRC?

Answer: Cyclic Redundancy Check is an error-detection technique that treats a bit string as a polynomial over GF(2). The sender divides the augmented data by a generator polynomial using modulo-2 arithmetic and appends the remainder.

Q3. What is ASCII?

Answer: ASCII is a character-encoding standard that represents basic English letters, digits, punctuation, and control characters using numeric codes. In an 8-bit representation, each character is represented by eight bits.

Q4. What is a generator polynomial?

Answer: The generator polynomial defines the divisor used in CRC. In bit-string form, its leading and trailing bits are normally 1. If its length is n bits, the CRC remainder has n–1 bits.

Q5. Why are n–1 zeros appended before division?

Answer: The sender appends n–1 zeros to reserve space for the CRC remainder. The modulo-2 division produces the remainder that is then placed in those reserved bit positions.

Q6. What is modulo-2 division?

Answer: Modulo-2 division uses XOR instead of ordinary subtraction. There are no carries or borrows: 0 XOR 0 = 0, 1 XOR 0 = 1, 0 XOR 1 = 1, and 1 XOR 1 = 0.

Q7. How is the transmitted CRC frame generated?

Answer: Convert the message to bits, append n–1 zeros, divide by the generator using XOR long division, take the final n–1 bits as the remainder, and append that remainder to the original data.

Q8. How does the receiver check a CRC frame?

Answer: The receiver divides the complete received codeword by the same generator polynomial. A zero remainder means no error was detected; a non-zero remainder indicates a detected error.

Q9. Can CRC correct an error?

Answer: CRC is an error-detection method, not an error-correction method. A system may request retransmission using an ARQ protocol, or use a separate error-correcting code if correction is required.

Q10. What is a burst error?

Answer: A burst error affects a sequence of bits over a span of a frame; not every bit in that span must be wrong. CRC polynomials are selected to detect important classes of burst errors.

Q11. Why is CRC stronger than a single parity bit?

Answer: A parity bit detects only odd numbers of bit flips in the protected group and misses even-numbered flips. A well-chosen CRC detects many more error patterns, including all single-bit errors and many burst errors.

Q12. What is the Hamming distance connection?

Answer: The minimum Hamming distance of a code is the smallest number of bit positions in which two valid codewords differ. It helps characterize error detection and correction capability; CRC performance depends on the chosen generator and frame length.

Q13. What happens if the receiver uses a different generator polynomial?

Answer: The sender and receiver must use the same agreed generator. A mismatch can produce a non-zero remainder even for an uncorrupted frame or fail to perform the intended check.

Q14. What is the CRC remainder called?

Answer: It is commonly called the CRC, checksum, or frame check sequence (FCS), depending on the context. It is redundant check information appended to the data.

Q15. Does a zero CRC remainder prove the data is certainly correct?

Answer: No. It means no error was detected by that CRC check. Some error patterns can produce a zero syndrome/remainder, so CRC does not provide an absolute mathematical guarantee against every possible corruption.

Experiment 4 — Go-Back-N and Selective Repeat Sliding Window Protocols

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. What is the sliding window protocol?

Answer: A sliding window protocol allows multiple frames to be sent before acknowledgements arrive, while limiting the number of outstanding unacknowledged frames. It provides flow control and can support error recovery.

Q2. What is ARQ?

Answer: Automatic Repeat reQuest is an error-control method in which a sender retransmits data when acknowledgements are missing or negative acknowledgements indicate a problem.

Q3. What is Go-Back-N ARQ?

Answer: Go-Back-N allows the sender to transmit a window of frames. The receiver accepts only the next expected frame and discards out-of-order frames. When a frame is lost, the sender retransmits that frame and subsequent outstanding frames.

Q4. What is Selective Repeat ARQ?

Answer: Selective Repeat allows the receiver to buffer correctly received out-of-order frames and acknowledge frames individually. The sender retransmits only frames that are missing or damaged.

Q5. What is the main difference between GBN and SR?

Answer: GBN uses cumulative acknowledgements and may retransmit several frames after one loss. SR uses individual acknowledgements and selective retransmission, reducing unnecessary retransmissions but requiring more buffering and logic.

Q6. What is a sender window?

Answer: The sender window is the range of sequence-numbered frames that may be sent without waiting for further acknowledgements. As acknowledgements arrive, the window advances or slides forward.

Q7. What is a receiver window?

Answer: The receiver window describes which sequence numbers the receiver is prepared to accept. In GBN it is effectively one frame; in SR it can cover multiple sequence numbers.

Q8. What is a cumulative ACK?

Answer: A cumulative ACK confirms receipt of all frames up to a particular point in sequence, typically indicating the next frame expected. It is characteristic of Go-Back-N.

Q9. Why does GBN discard out-of-order frames?

Answer: The standard GBN receiver accepts only the next expected frame and does not buffer later frames. This simplifies receiver logic but means those later frames may need retransmission.

Q10. Why does SR need more buffer space?

Answer: SR retains correctly received frames that arrive out of order until missing earlier frames arrive. It therefore needs storage and tracking for multiple outstanding sequence numbers.

Q11. What happens when an ACK is lost in GBN?

Answer: The sender may time out and retransmit the outstanding window starting from the oldest unacknowledged frame. Duplicate frames are recognized by sequence number and the receiver's cumulative ACK state.

Q12. What is a timeout?

Answer: A timeout is the sender's indication that an expected acknowledgement has not arrived within the configured time. It triggers retransmission or another recovery action.

Q13. What is the maximum window size for GBN with m-bit sequence numbers?

Answer: For a sequence-number space of 2^m, the usual maximum sender window for Go-Back-N is 2^m – 1. This avoids ambiguity between old and new frames after sequence numbers wrap around.

Q14. What is the maximum window size for Selective Repeat with m-bit sequence numbers?

Answer: The usual maximum sender and receiver window size is 2^(m–1), half the sequence-number space. This prevents ambiguity between old outstanding frames and new frames after wraparound.

Q15. Which protocol is more efficient on a noisy link?

Answer: Selective Repeat is generally more bandwidth-efficient when losses occur because it retransmits only missing frames. Its cost is greater receiver buffering and protocol complexity.

Experiment 5 — TCP Socket Programming: Hello, File Transfer, Calculator

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. What is a socket?

Answer: A socket is an endpoint used by an application to communicate over a network. It is commonly identified by an IP address, transport protocol, and port number.

Q2. What is TCP?

Answer: Transmission Control Protocol is a connection-oriented transport protocol that provides a reliable, ordered byte stream between endpoints. It uses acknowledgements, sequence numbers, retransmission, and flow/congestion control.

Q3. What is the TCP three-way handshake?

Answer: The connection is established through SYN, SYN-ACK, and ACK segments. This synchronizes sequence numbers and establishes the TCP connection before application data is exchanged.

Q4. What is the difference between TCP and UDP?

Answer: TCP is connection-oriented and provides reliable ordered byte-stream delivery. UDP is connectionless and sends independent datagrams without built-in delivery, ordering, or retransmission guarantees.

Q5. What is the purpose of socket()?

Answer: socket() creates a communication endpoint and returns a socket descriptor or handle. In POSIX C++ socket code, AF_INET selects IPv4 and SOCK_STREAM selects TCP.

Q6. What does bind() do?

Answer: bind() associates a server socket with a local IP address and port so clients can reach the service at that endpoint.

Q7. What does listen() do?

Answer: listen() marks a TCP socket as a passive listening socket and sets a queue limit for pending connection requests.

Q8. What does accept() do?

Answer: accept() accepts an incoming TCP connection and returns a new connected socket for communicating with that client. The original listening socket remains available for more clients.

Q9. What does connect() do?

Answer: connect() is called by a TCP client to initiate a connection to the server's IP address and port. It normally completes after the TCP handshake succeeds.

Q10. Why is a server started before the client?

Answer: The server must create its listening socket and wait for incoming connections. If the client connects before a server is listening, the connection attempt may be refused.

Q11. What is localhost or 127.0.0.1?

Answer: It is the IPv4 loopback address that refers to the same machine. It is useful for testing client and server programs on one computer without using the LAN.

Q12. What is a port number?

Answer: A port number identifies an application endpoint on a host. TCP and UDP use 16-bit port numbers so multiple services can share one IP address.

Q13. Why should file transfer use binary mode?

Answer: Binary mode preserves every byte without text newline conversion. This is necessary for arbitrary files such as images, PDFs, archives, and executables.

Q14. Does one send() always match one recv() in TCP?

Answer: No. TCP is a byte stream and does not preserve application message boundaries. send() and recv() may process partial amounts, so robust applications use loops and framing such as fixed lengths, delimiters, or length prefixes.

Q15. How does the calculator client-server program work?

Answer: The client sends an operation and operands using an agreed message format. The server parses the request, validates the operator and operands, performs the calculation, and sends the result back. Division by zero must be handled explicitly.

Experiment 6 — UDP Socket File Transfer

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. What is UDP?

Answer: User Datagram Protocol is a connectionless transport protocol that sends independent datagrams. It has low overhead but does not guarantee delivery, ordering, or duplicate protection.

Q2. What is a datagram?

Answer: A datagram is a self-contained packet of data sent using a connectionless protocol such as UDP. Each datagram carries addressing information and is delivered independently.

Q3. Why is UDP called connectionless?

Answer: A UDP sender can transmit a datagram to a destination address and port without first establishing a transport connection or performing a handshake.

Q4. Which socket type is used for UDP?

Answer: In POSIX socket programming, SOCK_DGRAM selects datagram sockets. In Java, DatagramSocket and DatagramPacket are commonly used.

Q5. What is sendto() used for?

Answer: sendto() sends a datagram to a specified destination address and port. The destination is supplied with the call rather than relying on a TCP-style accepted connection.

Q6. What is recvfrom() used for?

Answer: recvfrom() receives a datagram and can return the sender's address and port. A server can use this information to identify where a reply should be sent.

Q7. Why does a UDP file transfer divide a file into chunks?

Answer: UDP datagrams have size limits and large IP packets may fragment. Sending a file in manageable chunks makes transfer practical and allows each datagram to carry a portion of the file.

Q8. Does UDP guarantee that all file chunks arrive?

Answer: No. UDP itself provides no delivery guarantee. A datagram may be lost, duplicated, or delivered out of order; a basic teaching program may not detect or recover from all these cases.

Q9. Why are sequence numbers useful in UDP file transfer?

Answer: Sequence numbers let the receiver identify each chunk's position, detect missing or duplicate chunks, and reassemble data in the correct order.

Q10. What are ACKs and retransmissions?

Answer: An ACK is a message confirming receipt of data. If an ACK does not arrive before a timeout, the sender may retransmit the missing chunk. These mechanisms add reliability above UDP.

Q11. Why is an end marker potentially unsafe?

Answer: If the end-marker datagram is lost, the receiver may wait indefinitely; if file data happens to contain the marker, the receiver may stop early. A length field or explicit transfer protocol is safer.

Q12. What is the difference between TCP file transfer and a simple UDP file transfer?

Answer: TCP supplies reliable ordered stream delivery, so the application can send bytes and detect connection closure. A simple UDP transfer must handle packet loss, ordering, duplication, and completion itself if reliability is required.

Q13. What is the purpose of a receiver port?

Answer: The receiver binds to a known UDP port so the sender can address datagrams to that service. Both sides must agree on the destination port.

Q14. Can UDP transfer binary files?

Answer: Yes. UDP carries bytes, so a program can transfer any file type if it preserves exact byte values, uses suitable chunk sizes, and correctly reconstructs the received byte stream.

Q15. Where is UDP commonly used?

Answer: UDP is used by DNS, DHCP, real-time voice/video, online games, and other applications where low latency or application-specific reliability is preferred. Some modern protocols also build reliability on top of UDP.

Experiment 7 — TCP Peer-to-Peer and Multiuser Chat

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. Why is TCP suitable for a chat application?

Answer: TCP provides reliable, ordered delivery of a byte stream, which is useful for text messages that should arrive in order. The application still needs message framing because TCP does not preserve message boundaries.

Q2. What is peer-to-peer chat?

Answer: Peer-to-peer chat describes direct communication between two participants without a central message-broadcast server in the data path. A practical implementation may still designate one peer to listen for a connection.

Q3. What is multiuser chat?

Answer: A multiuser chat server accepts connections from multiple clients and distributes messages to the connected participants according to the application's broadcast rules.

Q4. What is the role of the chat server?

Answer: The server listens for client connections, accepts each client, receives messages, and forwards or broadcasts them to other connected clients.

Q5. Why is multithreading useful in a chat server?

Answer: A server can dedicate a thread to each connected client or use another concurrency model. This allows multiple clients to send and receive messages without one client blocking all others.

Q6. What is a race condition in a chat server?

Answer: A race condition occurs when concurrent threads access shared state in an uncontrolled way, potentially corrupting the client list or message output. Synchronization such as mutexes can protect shared data.

Q7. Why should disconnected clients be removed from the client list?

Answer: Removing disconnected sockets prevents attempts to send to invalid connections, avoids resource leaks, and keeps broadcasts limited to active clients.

Q8. What is broadcasting?

Answer: Broadcasting in a chat application means sending a received message to multiple connected clients. This is application-level fan-out, not necessarily an Ethernet broadcast.

Q9. What is a TCP port used for in chat?

Answer: The server listens on a configured port, and clients connect to that port. The IP address identifies the host and the port identifies the chat service.

Q10. What is graceful disconnection?

Answer: Graceful disconnection means the client signals that it is finished, closes its socket, and the server detects the end of the stream, cleans up resources, and updates the active-client list.

Q11. Why is message framing required over TCP?

Answer: TCP transports a continuous byte stream. A receiver may read part of one message or multiple messages together, so programs need delimiters, fixed-size records, or length-prefixed messages.

Q12. What is the difference between send() and recv()?

Answer: send() writes bytes to a connected socket and recv() reads bytes from it. Both can process fewer bytes than requested, so robust code checks return values and loops as needed.

Q13. How can a server support multiple clients?

Answer: It can accept connections in a loop and handle each client with a separate thread, a thread pool, or non-blocking I/O with an event loop. The choice depends on scale and complexity.

Q14. What happens if one client sends a very long message?

Answer: The server should enforce a message-size limit, read complete framed messages safely, and avoid unbounded memory use. It should reject or truncate invalid input according to the protocol.

Q15. What security concerns exist in a chat application?

Answer: Applications should validate input, authenticate users if needed, protect transport with TLS on untrusted networks, avoid logging secrets, and limit message sizes and connection rates.

Experiment 8 — UDP Peer-to-Peer and Multiuser Chat

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. How does UDP chat differ from TCP chat?

Answer: UDP chat sends independent datagrams and does not establish a connection. It has lower transport overhead but messages can be lost, duplicated, or reordered; TCP provides reliable ordered delivery.

Q2. What information is needed to send a UDP message?

Answer: The sender needs the destination IP address and UDP port, as well as the message bytes. Each datagram is addressed independently.

Q3. What is DatagramSocket in Java?

Answer: DatagramSocket is a Java class used to send and receive UDP datagrams. It can be bound to a local port and used with DatagramPacket objects.

Q4. What is DatagramPacket in Java?

Answer: DatagramPacket represents the data and addressing information for a UDP datagram. For sending it includes destination address and port; for receiving it provides a buffer and then exposes the sender details.

Q5. How does a UDP server learn the sender's address?

Answer: A receive operation such as recvfrom() or Java's receive(DatagramPacket) provides the source IP address and source port of the datagram. The server can use those details to reply.

Q6. Why does UDP not use listen() or accept()?

Answer: Those calls are part of the TCP connection-oriented server model. UDP has no connection handshake; a server binds a datagram socket and receives packets directly.

Q7. How can a UDP multiuser chat server broadcast a message?

Answer: The server records or otherwise knows the IP/port endpoint of each participant, receives a message, and sends a copy as a separate datagram to each other participant.

Q8. What happens if a UDP chat message is lost?

Answer: The receiving participant may never see it because UDP does not automatically retransmit. If reliable chat is required, the application must add acknowledgements, sequence numbers, and retransmission or use TCP.

Q9. Can UDP messages arrive out of order?

Answer: Yes. Datagrams are independent and may take different paths or experience different delays. Sequence numbers can help the application detect and reorder messages.

Q10. What is the maximum UDP payload size?

Answer: The UDP length field is 16 bits, but practical payload size is constrained by IP and network MTU. Large datagrams may fragment; applications should generally keep datagrams small to avoid fragmentation.

Q11. Why is a timeout useful in UDP applications?

Answer: A timeout lets an application stop waiting indefinitely for a response. It can trigger retry logic or report that a peer is unavailable.

Q12. Is UDP faster than TCP?

Answer: UDP has less built-in protocol overhead, but it is not automatically faster in every application. Performance depends on network conditions, payload size, loss, and how much reliability the application must implement.

Q13. What is a UDP port?

Answer: A UDP port identifies an application endpoint on a host. It is separate from a TCP port namespace even when the numeric value is the same.

Q14. What is the difference between unicast and broadcast?

Answer: Unicast sends a packet to one destination. Broadcast sends to all hosts in a broadcast domain where supported. A chat server sending one datagram to each registered client is application-level unicast fan-out.

Q15. What limitations should be mentioned during the practical?

Answer: UDP does not guarantee delivery, order, or uniqueness. A simple demonstration may show message exchange, but it should not be described as reliable unless it implements application-level acknowledgements, sequencing, and recovery.

Experiment 9 — Installation and Configuration of Web Server

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. What is a web server?

Answer: A web server is software that accepts HTTP requests and returns responses such as HTML pages, images, stylesheets, or application data. Apache HTTP Server is one widely used web-server implementation.

Q2. What is the client-server model in web browsing?

Answer: The browser acts as a client and sends an HTTP request to a server. The server processes the request and returns an HTTP response containing a status code, headers, and usually a body.

Q3. What is HTTP?

Answer: Hypertext Transfer Protocol is an application-layer protocol used for request-response communication on the web. HTTP commonly uses TCP port 80; HTTPS uses HTTP over TLS, commonly on port 443.

Q4. What is Apache HTTP Server?

Answer: Apache HTTP Server is open-source web-server software that can serve static content and support dynamic application configurations through modules and integrations.

Q5. What is the document root?

Answer: The document root is the directory from which the server serves website files. On a typical Ubuntu Apache installation, the default document root is /var/www/html.

Q6. What is index.html?

Answer: index.html is a common default directory index file. When a browser requests a directory URL, Apache may serve the configured index file from that directory.

Q7. How can you check whether Apache is running on Ubuntu?

Answer: Use `sudo systemctl status apache2` to inspect the service. You can also request the local site with a browser at http://localhost or use `curl -I http://localhost` to inspect the HTTP response headers.

Q8. What does systemctl do?

Answer: systemctl is the command-line interface for managing services under systemd. It can start, stop, restart, enable at boot, and report the status of Apache.

Q9. What is the difference between start and enable?

Answer: start launches the service now. enable configures the service to start automatically during future system boots. One does not imply the other.

Q10. How can another computer access the web server?

Answer: The server and client must have network connectivity. Find the server's LAN IP address, ensure Apache listens on the relevant interface and the firewall permits inbound TCP port 80, then browse to http://server-IP.

Q11. What is port 80?

Answer: TCP port 80 is the standard port for unencrypted HTTP. HTTPS normally uses TCP port 443, though services can be configured on other ports.

Q12. What is the purpose of the Apache access log?

Answer: The access log records incoming requests, commonly including client address, request line, status code, response size, and timestamp. It helps diagnose which requests reached the server.

Q13. What is the purpose of the Apache error log?

Answer: The error log records server errors and diagnostic messages such as configuration problems, permission errors, and failures while serving requests.

Q14. What is a firewall?

Answer: A firewall filters network traffic according to rules. For a basic HTTP lab, inbound TCP port 80 may need to be allowed; only necessary ports should be exposed.

Q15. What does HTTP status code 200 mean?

Answer: 200 OK indicates that the request succeeded. Other common statuses include 301/302 for redirection, 403 Forbidden for access denied, and 404 Not Found when a requested resource is unavailable.

Experiment 10 — Wireshark Packet Capture and Filter Expressions

Viva focus: Core concepts, working principles, configuration/program behaviour, testing, and common follow-up questions.

Q1. What is Wireshark?

Answer: Wireshark is a network protocol analyzer that captures packets from a selected interface and decodes protocol headers and available payloads for troubleshooting and learning.

Q2. What is packet capture?

Answer: Packet capture records network frames or packets observed on an interface. The captured data can be saved in a file such as .pcapng for later analysis.

Q3. What is the difference between a capture filter and a display filter?

Answer: A capture filter is applied before or during capture and controls which packets are recorded, commonly using BPF syntax. A display filter is applied after capture and only changes which captured packets are shown.

Q4. Give an example of a capture filter.

Answer: A capture filter such as `host 192.0.2.10` records traffic to or from that host. `tcp port 443` restricts capture to TCP traffic using port 443.

Q5. Give an example of a display filter.

Answer: `tcp` displays TCP packets. `http` displays packets Wireshark recognizes as HTTP. `ip.addr == 192.0.2.10` displays packets whose source or destination IPv4 address matches that address.

Q6. What does the SYN flag indicate?

Answer: SYN is used to synchronize sequence numbers and initiate a TCP connection. A typical first handshake packet has SYN set and ACK not set.

Q7. What does the PSH flag indicate?

Answer: PSH (Push) asks the receiving TCP implementation to pass buffered data to the receiving application promptly. It does not mean that a complete application message is necessarily present in one segment.

Q8. What does the RST flag indicate?

Answer: RST (Reset) is used to reset or abort a TCP connection, for example when a segment arrives for a connection that does not exist or a connection must be terminated abnormally.

Q9. What is the TCP three-way handshake visible in Wireshark?

Answer: The usual sequence is client SYN, server SYN-ACK, and client ACK. Inspecting sequence and acknowledgement numbers helps confirm the connection setup.

Q10. What is a packet's source and destination IP address?

Answer: The source IP identifies the sending host at the IP layer and the destination IP identifies the intended receiving host. Routers use destination IP addresses to forward packets.

Q11. What are source and destination ports?

Answer: TCP/UDP port numbers identify the sending and receiving application endpoints. They help distinguish multiple services and client connections on the same host.

Q12. Why might an HTTP filter show few or no packets when visiting Facebook?

Answer: Modern websites such as Facebook generally use HTTPS. The captured traffic is encrypted TLS over TCP or QUIC over UDP, so Wireshark may not decode it as readable HTTP. A display filter for `http` may therefore show little or no application payload.

Q13. What can be observed when traffic is encrypted with HTTPS?

Answer: Without decryption keys, packet metadata such as IP addresses, ports, packet sizes, timing, TCP/TLS handshakes, and some visible handshake metadata may be available, but page contents and login credentials should not be readable.

Q14. What is the purpose of Statistics → Conversations?

Answer: The Conversations view summarizes traffic exchanged between endpoint pairs, including packet and byte counts. It helps identify which hosts communicated and how much traffic was exchanged.

Q15. What is the difference between a packet and a frame?

Answer: A frame is the data-link-layer unit transmitted on a local link, such as an Ethernet frame. A packet commonly refers to a network-layer unit such as an IP packet carried inside a frame.

Final Practical-Day Rapid Revision

Experiment 1	Same subnet + unique IPs; switch forwards by MAC; ping uses ICMP; Windows sharing commonly uses SMB.
Experiment 2	Router connects different networks; configure interfaces, gateways, and routes; access point connects wireless clients.
Experiment 3	Append generator-length minus one zeros; XOR divide; append remainder; receiver checks for zero remainder.
Experiment 4	GBN retransmits from loss onward; SR retransmits only missing frames and buffers out-of-order frames.
Experiment 5	TCP server: socket → bind → listen → accept; client: socket → connect; exchange bytes with send/recv.
Experiment 6	UDP uses datagrams and sendto/recvfrom; no built-in delivery, ordering, or retransmission guarantee.
Experiment 7	TCP chat needs framing and concurrent client handling for multiuser broadcast.
Experiment 8	UDP chat sends each datagram to a client IP/port; application must handle loss if reliability is required.
Experiment 9	Apache serves files from its document root; check service status, localhost, LAN IP, firewall, and logs.
Experiment 10	Capture filters act before capture; display filters act after; SYN starts, PSH pushes, RST resets; HTTPS payload is encrypted.

Examiner tip: When asked about an experiment, begin with its aim, explain the key concept, describe the steps or program flow, then state how you verified the output. If a limitation exists in a basic demonstration, acknowledge it accurately rather than claiming a feature the program does not implement.